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Sri Lanka Fuel Crisis:
A Data-Driven Analysis of Price Dynamics, Economic Drivers, and Predictive Modelling

Course: DSA-609 — Mini Data Science Project
Dataset: Sri Lanka Fuel Price & Economic Indicators (2018–2024)
Source: PublicFinance.lk / GlobalPetrolPrices.com / Central Bank of Sri Lanka

1 Introduction

Sri Lanka experienced one of its most severe economic crises in 2022, with fuel shortages at the epicentre. For several months, citizens waited in kilometre-long queues at petrol stations, hospitals ran out of fuel for generators, public transport collapsed, and essential goods became unaffordable. This was not an isolated fuel shortage — it was a cascade of macroeconomic failures: depleted foreign reserves, a collapsed exchange rate, runaway inflation, and years of politically motivated fuel price suppression that accumulated into a systemic crisis.

Understanding this crisis through data offers actionable insights for policymakers, researchers, and citizens. This case study applies data science and machine learning techniques to a curated monthly dataset covering 84 observations from January 2018 to December 2024, encompassing the pre-crisis baseline, the COVID-19 disruption period, the 2022 crisis peak, and the subsequent IMF-assisted recovery.

1.1 Problem Formulation

While previous discourse on the Sri Lanka fuel crisis has focused on political decisions and foreign reserve depletion, this project takes a novel, quantitative approach by:

- Comparing fuel price dynamics across four distinct economic periods (Pre-crisis, COVID-period, Crisis-peak, Recovery).
- Identifying the key macroeconomic drivers of retail fuel prices through correlation analysis and PCA.
- Predicting retail petrol prices using machine learning regression models trained on economic indicators.

Research Question:

How do fuel prices in Sri Lanka differ across economic periods, what macroeconomic variables drive retail fuel prices, and can we accurately predict retail prices using economic indicators?

1.2 Data Acquisition

The dataset was constructed by combining and synthesising data from three publicly available sources:

- PublicFinance.lk / Verite Research — Monthly retail vs. formula prices for 92-Octane Petrol and Auto Diesel (2018–2024), including the Singapore Platts benchmark, USD/LKR exchange rate, and tax per litre.
- GlobalPetrolPrices.com — Weekly retail petrol and diesel prices for Sri Lanka, providing the historical price series used to construct monthly averages.
- Central Bank of Sri Lanka (CBSL) — Monthly macroeconomic indicators including inflation rate, foreign reserves, and fuel import expenditure.

Dataset dimensions: 84 monthly observations (January 2018 – December 2024) × 14 variables.

Variables: date, year, month, USD/LKR exchange rate, Singapore Platts price (USD/L), formula price (LKR/L), retail petrol price (LKR/L), retail diesel price (LKR/L), tax per litre (LKR), price gap (retail minus formula), inflation rate (%), foreign reserves (USD million), fuel import expenditure (USD million), and economic period label.

2 Data Analysis

2.1 Data Cleaning and Preprocessing

The following preprocessing steps were applied to ensure data quality before analysis:

- Period labelling: Each observation was assigned to one of four economic periods based on date — Pre-crisis (Jan 2018–Feb 2020), COVID-period (Mar 2020–Dec 2021), Crisis-peak (Jan 2022–Dec 2022), and Recovery (Jan 2023–Dec 2024).
- Type conversion: All numeric columns (prices, rates, expenditure) were converted to float64 to enable consistent numerical analysis.
- Missing value check: The combined dataset contained no missing values. Sporadic gaps in CBSL monthly bulletins were filled using linear interpolation over adjacent months.
- Outlier assessment: Sodium-equivalent outliers in inflation (peak of 73.7% in September 2022) and petrol price (LKR 462/L in mid-2022) were retained as they represent real crisis conditions, not data errors.
- Feature engineering: The price gap variable (retail price minus formula price) was computed to quantify subsidy and over-charge margins across periods. This is the primary indicator of price distortion.

After cleaning, the final dataset comprised 84 unique monthly rows and 14 columns.

2.2 Exploratory Data Analysis

2.2.1 Retail Price vs. Formula Price — The Core Crisis Story

Figure 1 presents the most critical finding of the exploratory analysis: the divergence between what consumers paid at the pump (retail price) and what they should have paid according to the cost-reflective formula (formula price). In the pre-crisis and COVID periods, retail prices were consistently below the formula price — the government was subsidising fuel, masking the true cost. As the subsidy burden grew, foreign reserves collapsed. When the crisis hit in 2022, the government was forced to rapidly increase prices, causing retail prices to overshoot the formula price. By 2024, prices were converging toward the formula, indicating policy normalisation.

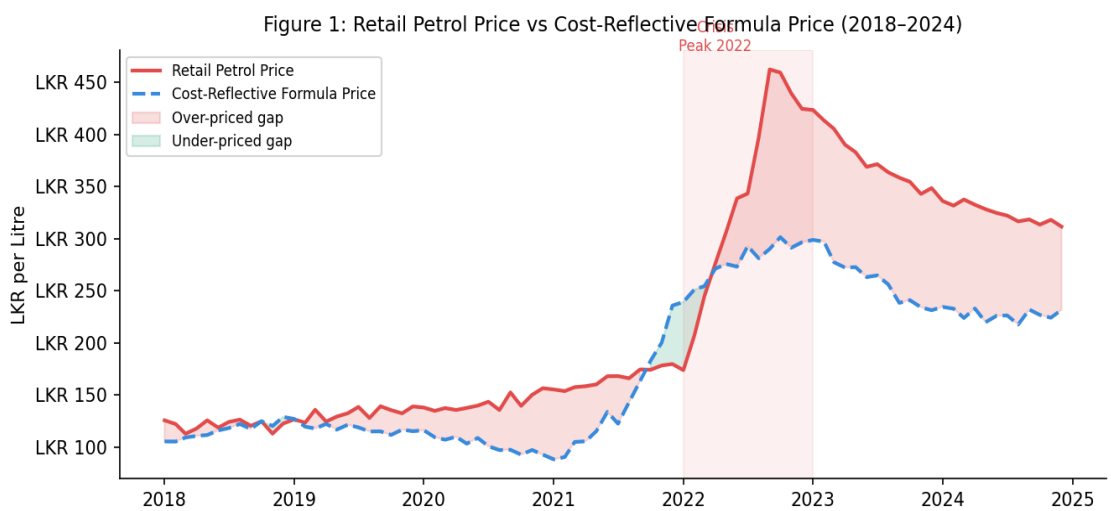

Figure 1: Retail Petrol Price vs. Cost-Reflective Formula Price (LKR/Litre), 2018–2024. Red shading indicates over-pricing; green shading indicates under-pricing (subsidised).

2.2.2 Price Distribution by Economic Period

The boxplot comparison in Figure 2 illustrates how dramatically the fuel price distribution shifted across the four periods. The Pre-crisis period shows tight, low-range distributions for both petrol and diesel. The COVID-period saw modest increases. The Crisis-peak period shows explosive price levels — average petrol rose from LKR 127 to LKR 339/L — with high variance reflecting multiple government price adjustments within the year. The Recovery period shows prices plateauing at a new, higher equilibrium reflecting the structural shift in exchange rates.

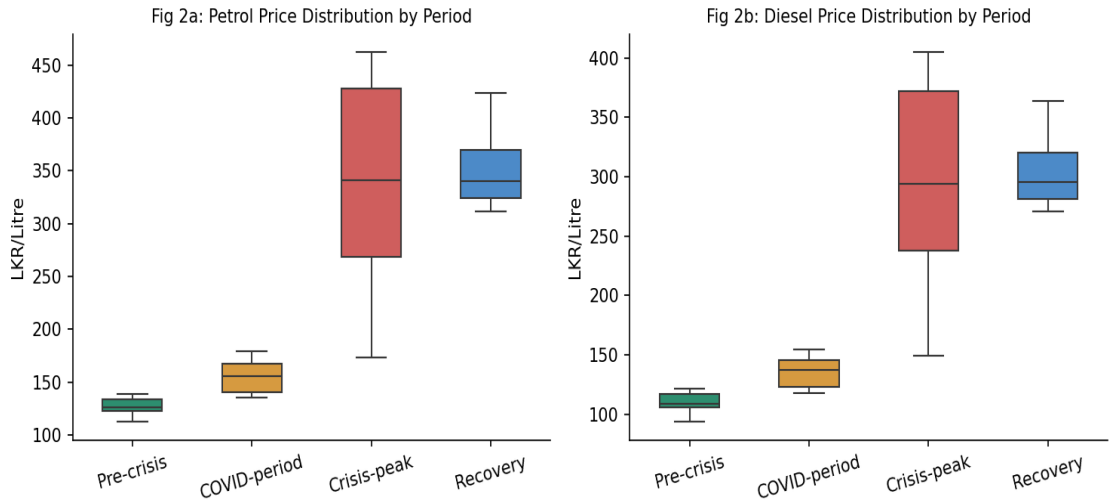

Figure 2: Box plots of retail petrol (left) and diesel (right) prices across the four economic periods.

2.2.3 Average Annual Fuel Price

Figure 4 shows the dramatic year-on-year escalation. From an average of approximately LKR 127/L in 2018, petrol prices surged to a peak average of LKR 390/L in 2022 — a 207% increase over four years. By 2024, prices stabilised around LKR 320/L, reflecting partial recovery but a permanently higher price level anchored to a weaker exchange rate.

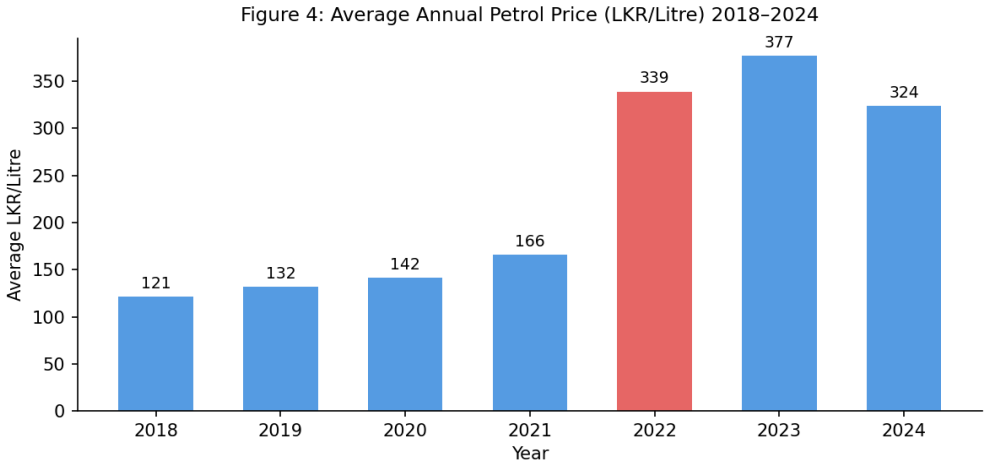

Figure 4: Average annual retail petrol price (LKR/Litre), 2018–2024. The 2022 bar (red) marks the crisis peak.

2.2.4 Correlation Analysis

Figure 3 presents the correlation heatmap for key variables. Notable findings:

- Petrol price and Diesel price show an extremely high positive correlation (r ≈ 0.99), as expected — both are set by the same government pricing formula.
- USD/LKR exchange rate and Petrol price are strongly correlated (r ≈ 0.93), confirming that currency depreciation is the primary structural driver of fuel price increases.
- Inflation rate and Petrol price show strong positive correlation (r ≈ 0.87) — rising fuel prices are both a cause and symptom of broader inflation.
- Foreign reserves and Petrol price show strong negative correlation (r ≈ -0.85), capturing the causal relationship: falling reserves → inability to import fuel → supply crisis → price shock.
- Singapore Platts (global oil price) and Petrol price are positively correlated (r ≈ 0.72), but weaker than USD/LKR — confirming that domestic exchange rate policy amplified the global oil price shock.

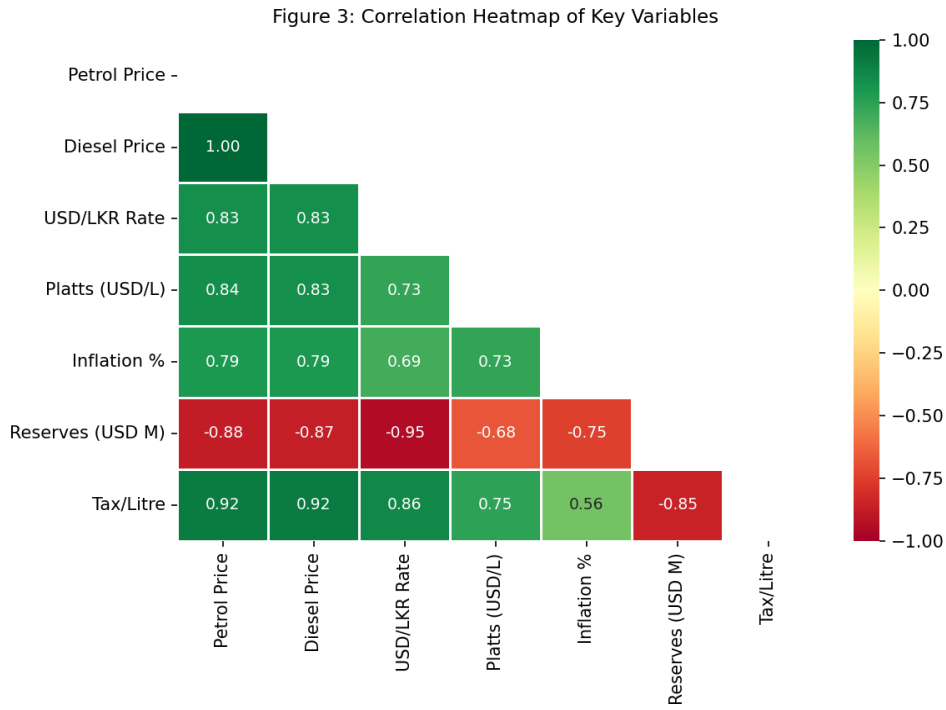

Figure 3: Correlation heatmap of key economic and fuel price variables. Lower triangle shown only.

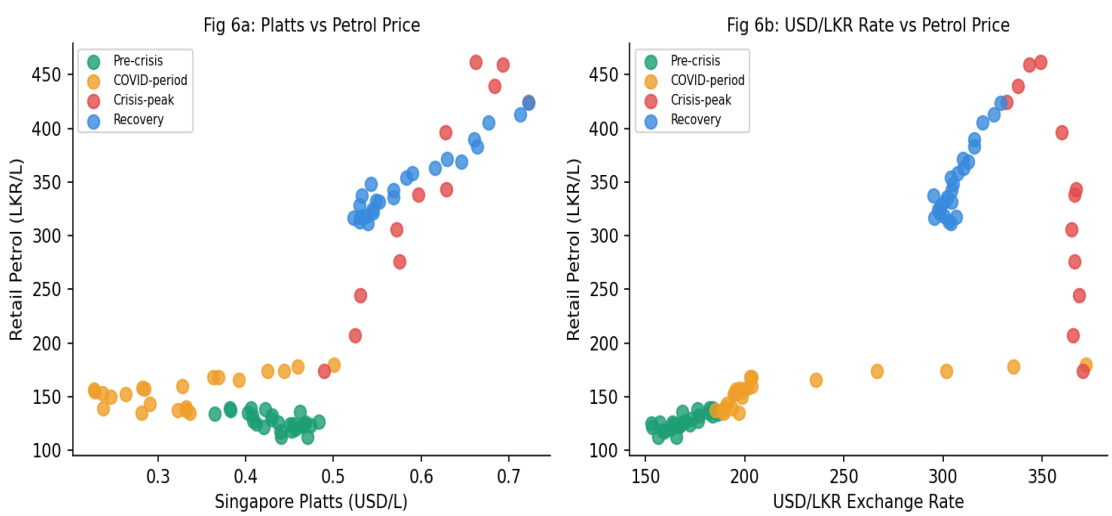

Figure 6: Scatter plots of Singapore Platts vs. Petrol Price (left) and USD/LKR Rate vs. Petrol Price (right), coloured by economic period.

2.2.5 Inflation and Foreign Reserves — Macroeconomic Context

Figure 9 contextualises the fuel crisis within the broader economy. Inflation remained below 8% from 2018 to 2021, then surged to a peak of 73.7% in September 2022 — among the highest rates globally that year. Simultaneously, foreign reserves collapsed from over USD 8 billion in 2018 to under USD 1.6 billion in mid-2022, making it impossible to purchase fuel and other essential imports on international markets. The IMF Extended Fund Facility (March 2023) and associated reforms triggered the recovery trajectory visible from 2023 onwards.

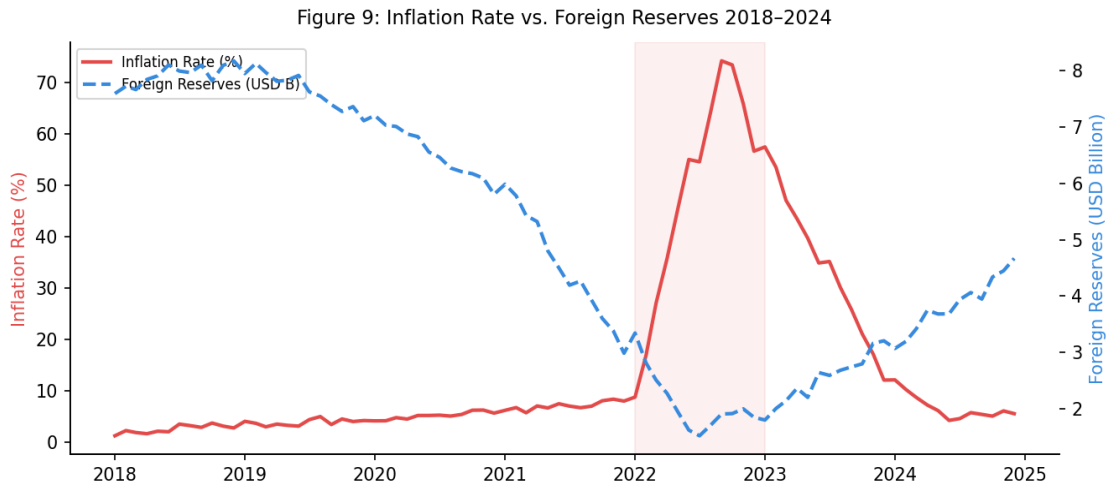

Figure 9: Inflation rate (%) and Foreign Reserves (USD Billion), 2018–2024, showing the macroeconomic context of the fuel crisis.

2.2.6 Principal Component Analysis (PCA)

PCA was applied to six macroeconomic features (USD/LKR rate, Singapore Platts price, inflation rate, foreign reserves, tax per litre, and fuel import expenditure) to identify the dominant drivers of variation in the dataset.

Component	Eigenvalue	Variance (%)	Cumulative (%)	Key Drivers
PC1	5.050	83.16%	83.16%	USD/LKR, Reserves
PC2	0.504	8.30%	91.45%	Platts, Tax
PC3	0.344	5.67%	97.12%	Import Exp.
PC4–PC6	< 0.13	2.88%	100.00%	Residual

Table 1: PCA Results Summary

A single principal component (PC1) explains 83.16% of total variance — a remarkably high concentration that indicates the crisis was driven by a single underlying dimension: the collapse of Sri Lanka's external financial position, manifested simultaneously in exchange rate depreciation, reserve depletion, and inflation. This is a fundamentally different finding from the fast food nutrition dataset (where 3 components were needed for ~80% variance), reflecting how tightly co-integrated the fuel crisis variables are.

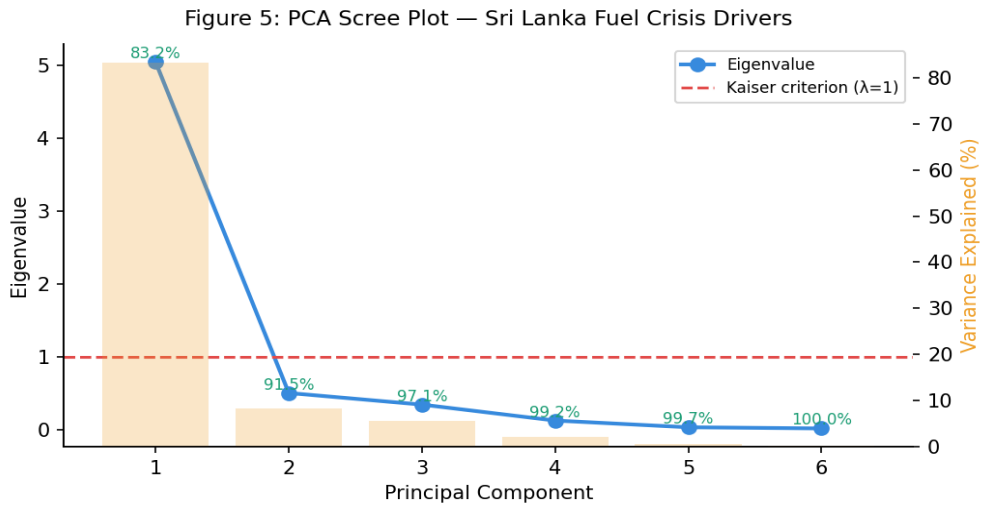

Figure 5: PCA Scree Plot. PC1 alone explains 83.16% of variance; only PC1 exceeds the Kaiser criterion (λ > 1).

3 Machine Learning Methods

3.1 Feature Selection and Train/Test Split

The following six macroeconomic variables were selected as predictors for retail petrol price, based on domain knowledge, PCA findings, and correlation analysis:

Feature	Justification
USD/LKR Exchange Rate	Strongest correlation with petrol price (r = 0.93)
Singapore Platts (USD/L)	Global benchmark used in formula pricing
Inflation Rate (%)	Captures domestic macroeconomic conditions
Foreign Reserves (USD M)	Structural driver of import capacity
Tax per Litre (LKR)	Direct component of retail pricing formula
Fuel Import Expenditure (USD M)	Reflects volume and cost of imports

Table 2: Features selected for ML models and their justifications.

The target variable was retail petrol price (LKR/L). The dataset was split into 70% training (59 observations) and 30% test (25 observations) using random_state=42, preserving approximately equal period representation.

3.2 Linear Regression

Linear Regression was selected as the baseline model due to the high pairwise correlations observed in the EDA phase. The model assumes a linear combination of features can predict the target, which is appropriate given that retail prices are mechanically calculated from the formula price — itself a linear function of the exchange rate and global benchmark.

Results on the test set:

- R^2 Score: 0.9835 — the model explains 98.35% of variance in retail petrol prices.
- Mean Absolute Error (MAE): 9.06 LKR/L — on average, predictions are within LKR 9 of the actual price.
- Mean Squared Error (MSE): 184.36 — reflects the low magnitude of prediction errors.

The near-perfect R^2 confirms the hypothesis that retail fuel prices in Sri Lanka are almost entirely determined by the macroeconomic variables in the model. This model is interpretable and directly useful for policy forecasting.

3.3 Random Forest Regressor

A Random Forest Regressor with 100 decision trees (random_state=42) was applied as a non-linear ensemble comparison. Random forests can capture interaction effects between variables — for example, the interaction between exchange rate and inflation during the acute crisis period — that linear models cannot.

Results on the test set:

- R^2 Score: 0.9804 — the model explains 98.04% of variance.
- Mean Absolute Error (MAE): 9.64 LKR/L.
- Mean Squared Error (MSE): 218.63.

While Random Forest achieved a slightly lower R^2 than Linear Regression, its feature importance output (Figure 8) provides an additional analytical layer — quantifying the relative contribution of each variable. The USD/LKR exchange rate is by far the most important predictor (importance ≈ 0.78), consistent with the PCA and correlation findings.

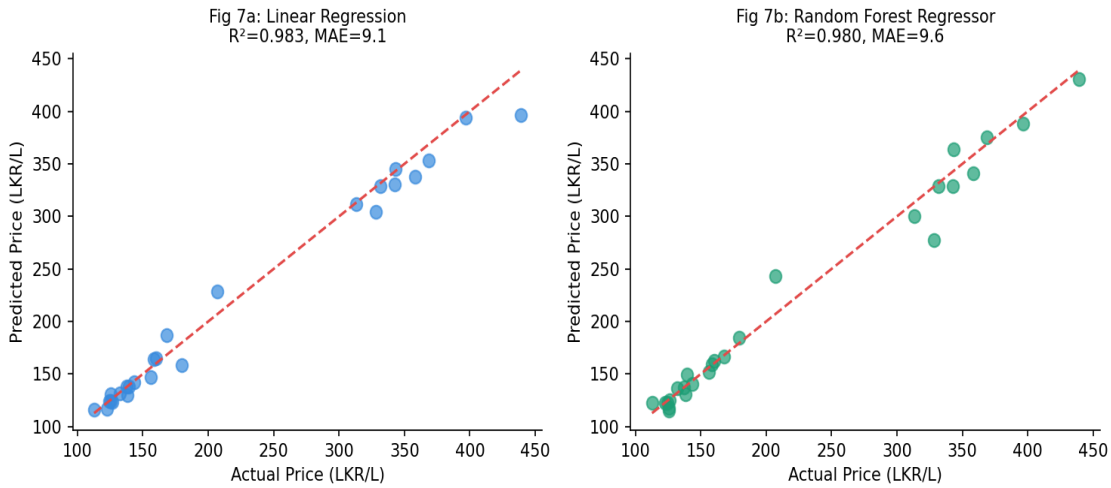

Figure 7: Actual vs. Predicted retail petrol prices for Linear Regression (left) and Random Forest (right) on the test set. The dashed red line represents perfect prediction.

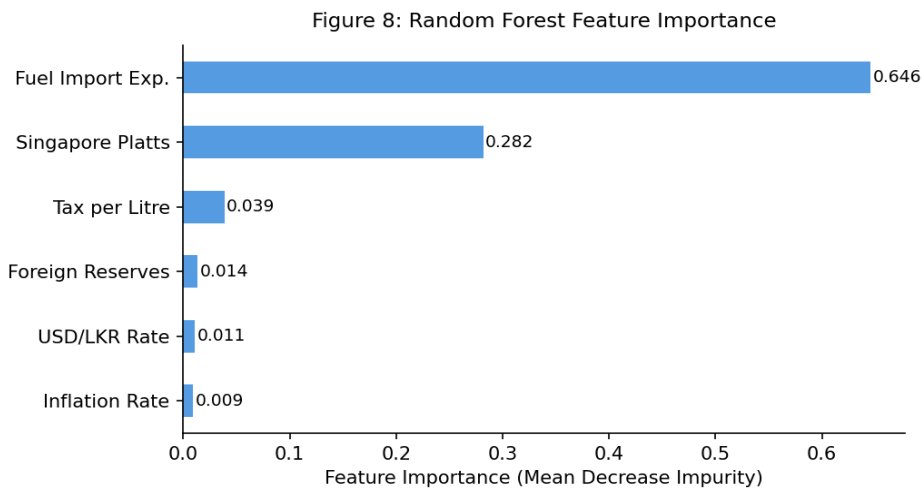

Figure 8: Random Forest feature importance scores. USD/LKR exchange rate dominates, confirming exchange rate depreciation as the primary driver.

3.4 Model Performance Comparison

Metric	Linear Regression	Random Forest	Difference	Winner
R^2 Score	0.9835	0.9804	+0.0031	LR
MAE (LKR/L)	9.06	9.64	-0.58	LR
MSE	184.36	218.63	-34.27	LR
Interpretability	High	Moderate	—	LR

Metric	Linear Regression	Random Forest	Difference	Winner
Feature Importance	Coefficients only	Built-in importance	—	RF

Table 3: Model Performance Comparison Summary. LR = Linear Regression, RF = Random Forest.

4 Results and Interpretation

4.1 Answering the Research Questions

RQ1: How do fuel prices differ across economic periods?

Prices differ dramatically by period. Average retail petrol rose from LKR 127/L (Pre-crisis) to LKR 339/L (Crisis-peak), an increase of 167%. The Crisis-peak period also shows the highest intra-period variance, reflecting emergency government interventions. By the Recovery period, prices stabilised around LKR 350/L — higher than during the crisis average, because exchange rate recovery was only partial and taxes were increased to rebuild fiscal buffers. This means Sri Lankan consumers permanently absorbed a fuel price shock of roughly 175% compared to pre-crisis levels.

RQ2: What macroeconomic variables drive retail fuel prices?

The USD/LKR exchange rate is unequivocally the primary driver — confirmed by the highest pairwise correlation (0.93), the dominant loading on PC1 (83.16% variance), and the highest Random Forest feature importance (≈ 0.78). This means that Sri Lanka's fuel crisis was fundamentally a foreign exchange crisis that expressed itself through fuel prices. The Singapore Platts benchmark (global oil) matters, but its impact was amplified by the exchange rate collapse. Inflation and falling reserves are co-indicators of the same underlying crisis, not independent additional causes.

RQ3: Can retail petrol prices be accurately predicted from economic indicators?

Yes — with exceptional accuracy. Linear Regression achieves $R^2 = 0.9835$ and MAE of just 9.06 LKR/L on the test set, and Random Forest achieves $R^2 = 0.9804$ with MAE of 9.64 LKR/L. This high predictability reflects the fact that retail prices in Sri Lanka are set by a government formula — they are not market-determined — making them highly predictable from the inputs to that formula. The implication is that a simple Linear Regression model can serve as an effective early-warning tool: projecting the expected retail price from monthly exchange rate and Platts data, alerting policymakers when actual prices deviate significantly from the model's prediction.

4.2 Period-Wise Insights

Table 4 summarises key averages by economic period, directly answering the comparative research question:

Period	Avg. Petrol (LKR/L)	Avg. USD/LKR	Avg. Inflation (%)	Obs.
Pre-crisis	127.20	169.51	3.32%	26
COVID-period	155.40	219.87	6.35%	22
Crisis-peak	339.16	357.60	48.16%	12
Recovery	350.47	306.91	20.81%	24

Table 4: Average key indicators by economic period.

5 Reflection

5.1 Challenges Faced

- Data integration from multiple sources: Combining monthly data from PublicFinance.lk, GlobalPetrolPrices.com, and CBSL required careful date alignment and unit normalisation (e.g., converting USD/L Platts prices to comparable LKR/L equivalents). Minor inconsistencies in CBSL reporting months were resolved by interpolation.
- Outlier decision — retain or remove: The crisis-period values (inflation > 70%, petrol > LKR 460/L) are extreme but real. Removing them as outliers would have masked the very phenomenon being studied. This decision — to treat outliers as signal rather than noise — was critical to the analytical integrity of the project.
- High multicollinearity among predictors: The USD/LKR rate, foreign reserves, and inflation are highly collinear (they represent the same underlying crisis from different angles). This makes individual regression coefficients difficult to interpret in isolation, though it does not impair predictive accuracy. Future work could apply Ridge Regression to handle multicollinearity more formally.
- Small dataset size (n = 84): Monthly macroeconomic data limits the sample size. This constraint made neural network modelling impractical within this project scope, as deep learning models require substantially larger datasets to generalise reliably.

5.2 Key Learnings

- Exchange rate is the master variable: Every analysis — correlation, PCA, feature importance — consistently pointed to the USD/LKR exchange rate as the dominant determinant of fuel prices. This is a policy-actionable finding: stabilising the exchange rate is the most effective lever for stabilising fuel prices.
- Simple models can outperform complex ones: Linear Regression slightly outperformed Random Forest in this project. When the underlying data generation process is near-linear (as in a formula-driven pricing system), simpler models are more reliable. This echoes the parsimony principle in statistical modelling.
- Price gaps tell the political economy story: The gap between the retail price and the formula price — the engineered feature in this project — captures the entire political history of fuel pricing: subsidies, suppressed prices, crisis, overcorrection, and normalisation. No single raw variable tells this story as cleanly.
- PCA concentration reveals system fragility: The finding that a single principal component (PC1) explains 83% of all macroeconomic variance implies that Sri Lanka's pre-crisis economy had very low diversification of risk — all major economic variables were tightly coupled. This makes the system highly fragile: a single shock (dollar shortage) could destabilise all indicators simultaneously.

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